Project Report: Student Academic Performance Predictor

Course: Fundamentals of AI and ML

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1. Problem Statement & Motivation

The Problem: In a competitive academic environment, students often struggle to balance study hours, attendance, and extra tutoring. Predicting a final exam score is traditionally reactive (based on the exam itself), which doesn't allow for early intervention.

Why it Maters: A predictive AI model can help students and faculty identify "at-risk" performance levels early in the semester. By understanding the weight of each factor (like attendance vs. self-study), students can optimize their routines for better results.

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1. Technical Approach

- Data Source: I utilized the StudentPerformanceFactors dataset, which contains real-world academic behavioural data.
- Features Selected: To move beyond simple correlation, I selected a multi - dimensional feature set:
- o Hours_Studied (Continuous)
- o Attendance (Continuous/Percentage)
- o Previous_Scores (Baseline Performance)
- o Tutoring_Sessions (Categorical/Numerical)

- Algorithm: I implemented Polynomial Regression (Degree 4). While Linear Regression assumes a constant rate of improvement, a Quar c

  (4th degree) model accounts for "Diminishing Returns"—where studying 80 hours might not be twice as effective as 40 hours due to burnout or saturation.

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2. Key Decisions & Architecture

- Preprocessing: Used pandas.dropna() to ensure the model was trained on high-quality, complete data points.
- Feature Expansion: Used PolynomialFeatures(degree=4) to transform the input space. This allows a linear solver to find a non-linear "best-fit" curve.
- Visualization on Strategy: Since humans cannot visualize 4D data, I used NumPy to create a synthetic "Average Student" baseline (fixing attendance and previous scores at their mean) to isolate and plot the impact of study hours in 2D.

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4. Challenges Faced & Solu ons

- The "FileNotFound" Error: Encountered issues with relative paths when running scripts from subfolders.

  Solution: Standardized the directory structure to keep the CSV and .py file in a single "Project" directory.

- Array Formattng Error: Experienced a TypeError when printing NumPy arrays in f-strings.

  Solution: Utilized index slicing [0] to extract scalar values from the prediction arrays for clean reporting.

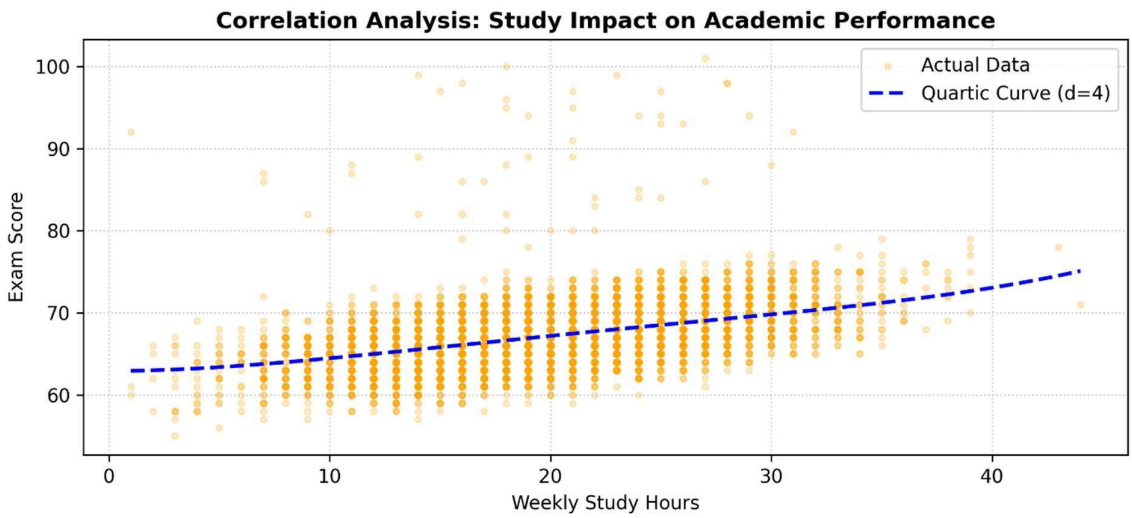
- Model Complexity: I initially tested Degree 7, but observed signs of Overfitting (the curve was chasing noise).

  Solution: Settled on Degree 4, which offered the best balance between accuracy (R^2) and generalizability.

5. Results & Analysis

- Degree 4 R^2 Score: 60.34
- Mean Squared Error (MSE): 6.16

Analysis: The Degree 4 model showed a significant improvement over the linear baseline. The lower MSE indicates that the "curved" mathematical approach more accurately represents the reality of student learning patterns.



6. Conclusion & Learnings

Through this BYOP, I learned that:

1. Data is never a straight line: Real-world behavior is curved and complex.
2. Mathema cs is the engine: Applying Linear Algebra (feature transformation) and Calculus (MSE optimization) is essential for modern AI.
3. Clean Code Matters: Professional formatting and error handling (try-except) are just as important as the algorithm itself for making a tool usable by others.